

$$\alpha_L = \frac{1}{N} \left(\frac{1 - e^{2\pi i (N\delta - L)}}{1 - e^{2\pi i (\delta - L/N)}} \right)$$

$$\alpha_0 = \frac{1}{N} \left(\frac{1 - e^{2\pi i N\delta}}{1 - e^{2\pi i \delta}} \right)$$

$$p_0 = \alpha_0^*$$

$$p_0 = \frac{1}{N^2} \frac{(2 - 2 \cos(2\pi N\delta))}{(2 - 2 \cos(2\pi \delta))} = \frac{1 - \cos 2\pi N\delta}{1 - \cos(2\pi \delta)}$$

$$(1 - \cos \theta) = 2 \sin^2 \theta/2 \rightarrow p_0 = \frac{2 \sin^2 \pi N\delta}{2 \sin^2(\pi \delta)} \frac{1}{N^2}$$

$$P_1 = \frac{1}{N^2} \alpha_1 = \frac{1}{N^2} \left(\frac{1 - e^{2\pi i(N\delta - 1)}}{1 - e^{2\pi i(\delta - 1/w)}} \right)$$

$$e^{i\theta} = \cos\theta + i\sin\theta \quad e^{2\pi i(N\delta - 1)} = e^{2\pi iN\delta} \cdot e^{-2\pi i}$$

$$e^{-2\pi i} = \cos(-2\pi) + i\sin(-2\pi) = \underline{1 + 0}$$

$$\delta - \frac{1}{N} = -\delta_1 = \delta - \frac{1}{N}$$

$$\alpha_1 = \frac{1}{N^2} \frac{1 - e^{2\pi iN\delta}}{1 - e^{-2\pi i\delta_1}} \quad p_1 = \alpha_1 \cdot \alpha_1^*$$

$$p_1 = \frac{1}{N^2} \frac{2 - 2\cos(2\pi N\delta)}{2 - 2\cos(2\pi\delta_1)} \rightarrow \frac{2(1 - \cos 2\pi N\delta)}{2(1 - \cos 2\pi N\delta_1)} \frac{1}{N^2}$$

$$1 - \cos\theta = 2\sin^2\theta/2 \rightarrow \frac{2\sin^2\pi N\delta}{2\sin^2(\pi\delta_1)} \frac{1}{N^2} = p_1$$

peak interpolation $\frac{P_1}{P_0} = r^2$ where $p_0 > p_1$

$$r^2 = \frac{1}{A^2} \frac{\sin^2 \pi N \delta}{\sin^2 \pi \delta_1} A^2 \frac{\sin^2 \pi \delta}{\sin^2 \pi N \delta} = \frac{\sin^2 \pi \delta}{\sin^2 \pi \delta_1} \quad \text{where}$$

$$\text{for small } x \quad \sin^2(x) = x^2 \rightarrow r^2 = \frac{\pi^2 \delta^2}{\pi^2 \delta_1^2}$$

$$\delta_1 = \frac{1}{N} - \delta \quad r^2 = \frac{\delta^2}{\delta_1^2} \quad r = \frac{\delta}{\delta_1} = \frac{\delta}{\frac{1}{N} - \delta}$$

$$r\left(\frac{1}{N} - \delta\right) = \delta \rightarrow r\frac{1}{N} - r\delta - \delta = 0$$

$$\cancel{r\frac{1}{N} - r\delta} + r\frac{1}{N} - \delta = 0 \quad \frac{rL}{N} - \delta(r+1) = 0$$

$$\frac{rL}{N} = \delta(r+1) \rightarrow \frac{rL}{N} \cdot \frac{1}{(r+1)} = \delta$$

$$\text{where } L=1 \quad \frac{r}{N(r+1)} = \delta$$